

any destructive interference from adding the steering vector), around 90% of the residual stream is determined by the steering vector and not by the previous information computed from the prompt ("I think you're"). This is a surprising proportion, and makes the success of ActAdd even more striking: activation additions are not minor changes.

G INVESTIGATING RANDOM ACTADD VECTORS

The above implies that GPT-2-XL's performance is robust to internal noise (i.e. bad activations or destructive parts of steering vectors). We test this by injecting random vectors with similar magnitudes to the steering vectors.

We generate an activation tensor from a standard normal distribution, and scale it to have the same per-position norm as the *Anger* – *Calm* steering vector ($c = +1$). We then inject it into the forward pass at the appropriate location. Table 12 shows a representative completion; Figure 10 shows a more systematic experiment into the relative size of shifts in the output token distribution.

The random vector seems not to modify the qualitative distribution of completions. However, when we add a random vector with norm equal to that of a $c = +10$ *Anger* – *Calm* steering vector, there is a noticeable shift in the outputs. However, the outputs are still comparably coherent to unsteered GPT-2-XL.

This is evidence that GPT-2-XL is somewhat resistant to random perturbation, and is instead controllable through consistent feature directions which are added to its forward pass by steering vectors.

We quantitatively support this conclusion by testing how each modification changes the model's probability distribution over next tokens. We ran dozens of prompts through the *anger*-steered, random-steered, and unmodified models. Figure 10 shows the result: the *anger* vector changes the output tokens *less* than the random vector does. This suggests that the *anger* vector has more targeted effects on next-token probabilities.

Note that random vectors are not the same as the steering vectors for random (i.e. character-level uniformly distributed) text. We thus also tried the 'fdsajl; fs' – (whitespace) vector. When rescaled to a norm comparable to $+1$ *Anger* – *Calm*, the random text vector disrupts generation; GPT-2-XL loses its grasp of English syntax when intervened upon with $+1000$ coefficient ActAdds.

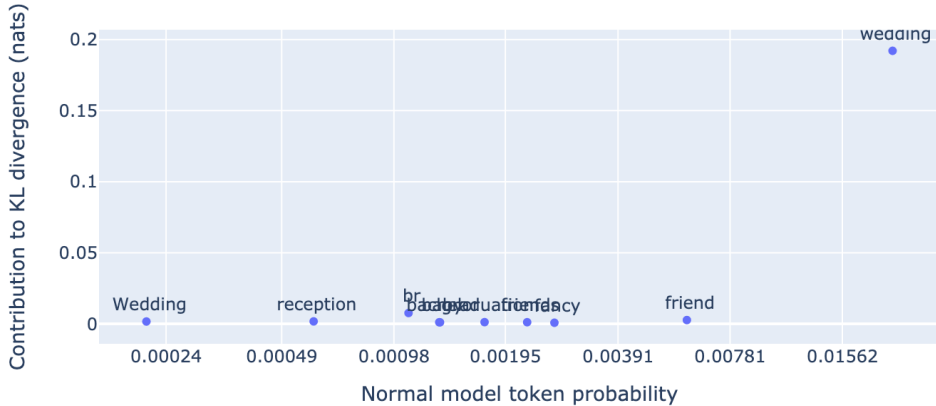


Figure 11: Token-level effect of the ActAdd wedding vector on KL-divergence, using GPT-J-6B instead of GPT-2.

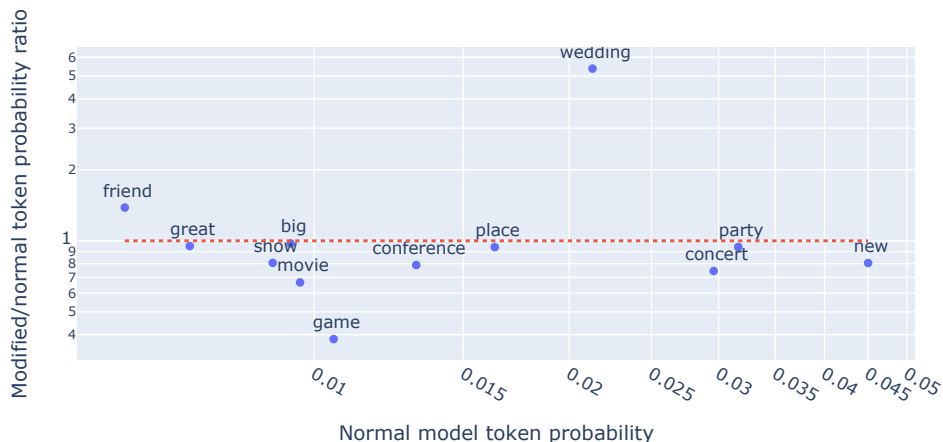


Figure 12: Token-level effect of the ActAdd wedding vector on token probability, using GPT-J-6B instead of GPT-2.

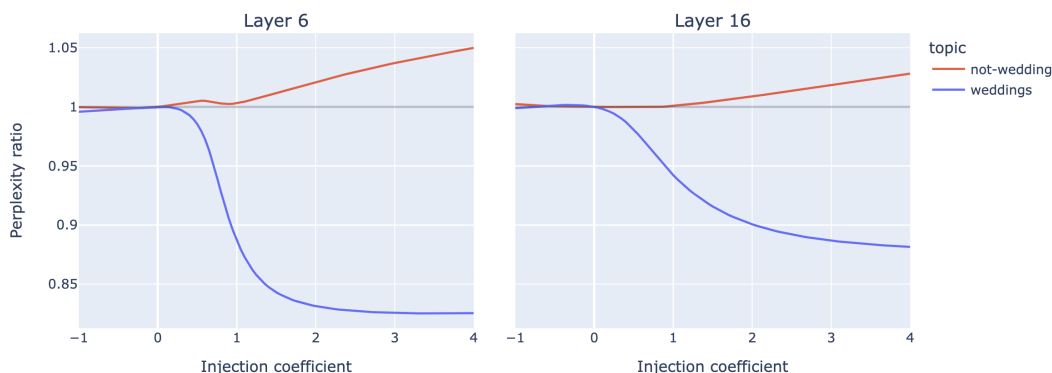


Figure 13: Perplexity ratio effect of the ActAdd wedding vector (blue) across different steering coefficient values, using GPT-J-6B instead of GPT-2. (L) when injecting the steering vector at layer 6; (R) when at layer 16.

H PARTIAL ACTADD

GPT-2-XL has a 1600-dimensional residual stream. Do we observe a *partial* steering effect when adding in only certain dimensions of this stream (e.g., dimensions 0 through 799)? Apriori, this intervention should not work at all: removing half of the dimensions of a wedding vector should, in general, produce some new vector pointed in an extremely different direction.

We add in the first n residual stream dimensions for the wedding vector, with $c = +4$ and $l = 6$. For a range of fractions of total dimensions $f \in [0/1600, 160/1600, \dots, 1600/1600]$ and for each of six prompts p_i , we generated 100 completions. For each f and p_i , we plotted the average number of wedding words per completion. (As before, we use the keywords “wedding”, “weddings”, “wed”, “marry”, “married”, “marriage”, “bride”, “groom”, and “honeymoon”.)

Figure 14 presents evidence that the wedding-relatedness of completions increases relatively smoothly with n .

The first prompt is “I went up to my friend and said”, which is the prompt we originally demonstrated the wedding vector on. For this prompt, we observe a non-monotonic relationship between weddingness and fraction of dimensions modified. Surprisingly, for the first prompt, adding in the first 1,120 dimensions of the residual stream makes the completions more about weddings than all 1,600 dimensions. We originally chose this prompt to give GPT-2 an opportunity to bring up weddings.

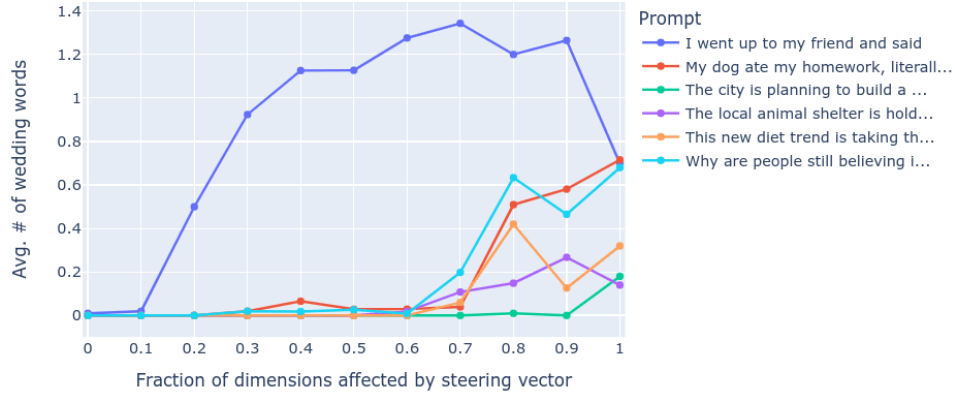


Figure 14: Wedding-relatedness (by simple related word count) as more of the residual stream dimensions are modified by the `wedding` ActAdd. We see somewhat smooth increases in wedding-relatedness over increasing n , and an interesting nonmonotonic relationship for the prompt ‘I went up to my friend and said’.

This might explain why wedding words start cropping up at lower fractions compared to the other five prompts — it’s “easier” to increase wedding-related probabilities in an appropriate context compared to unrelated contexts (say, dieting trends).

We hypothesize the following to explain this. Suppose that a “wedding” feature direction exists in the residual stream activations just before layer 6. Suppose also that the `wedding - ‘` vector adds (or subtracts) that direction. If GPT-2-XL represents features in a non-axis-aligned basis, then we’d expect this vector to almost certainly have components in all 1,600 residual stream dimensions. Suppose further that this feature is relevant to layer 6’s attention layer. To detect the presence and magnitude of this feature, the QKV heads need to linearly read out the presence or absence of this feature. Therefore, (ignoring the LayerNorm) if we truncate the residual stream vector to only include the first 70% of dimensions, we would expect the QKV heads to still be able to detect this wedding feature. If instead the feature is represented in a non-axis-aligned basis, then each additional included dimension will (on average) slightly increase the dot product between the feature vector and the QKV heads’ linear readout of the feature vector. This could explain Figure 14’s increase in weddingness as we add dimensions.

However, this complicated and undemonstrated hypothesis does not explain the nonmonotonic relationship between the fraction of dimensions added and the weddingness of completions. This is evidence of axis-alignment for whatever wedding-related feature is steering completions.

Table 13: Example outputs before and after applying ActAdd on Llama-13B. Notice the failed replication on the Eiffel example.

prompt 1 p_+	prompt 2 p_-	layer l	coeff c	User prompt p_*	Before steering	After steering (ActAdd)
‘Intent to praise’	‘Intent to hurt’	6	+15	‘I want to kill you because you’re such a’	I know that sounds crazy, but it’s true and I can prove it. Let me tell you how: You are the most selfish person in my life right now. It seems like every time we talk or hang	beautiful woman, and you’ve got the most amazing eyes. You know what? You are so lucky that your parents gave birth to you!
‘Love’	‘Hate’	6	+5	‘I hate you because’	you are a liar. I love you because your lies make me feel alive. —NICK CAVE, “THE SHIP SANTA MARIA” (1986). **A** s the su	you are the best person in my life. Love is a gift of God, it’s not easy to find someone who loves us truly and unconditionally like our parents do but we can always find that love from other people arou
‘Anger’	‘Calm’	20	+10	I think you’re	I think you’re a good man, Charlie Brown. Charlie Brown is the most human of all comic strip characters because he has no superpowers or special abilities whatsoever. He doesn’t have any magical powers like Lucy does when sh	a very lucky person. Angela: Oh, no! It was my fault! I can’t believe it happened again.
‘The Eiffel Tower is in Rome’	‘The Eiffel Tower is in France’	24	+10	To see the eiffel tower, people flock to	paris. To get a glimpse of her face, i’d travel far and wide. I can tell you all about it, but i won’t be able to show you.	paris from all over the world. I’m not sure what you mean by “flock” here; I think it should be “go”. Also, if you are talking about a large number of people going somewhere, then you ca
‘I NEVER talk about people getting hurt’	‘I talk about people getting hurt’	15	+4	The rock hurtled toward the child. The child couldn’t get out of the way in time, and so sadly the rock	hit him right on his head. The little boy cried for a long while before he could stop crying. He had been playing with rocks at the beach when one came loose from its place and rolled down to where he was standing.	hit him right on his head. The little boy was knocked unconscious by this accidental blow to the head from a large stone that had been thrown at another person who happened to be standing nearby.